

Recursive Offline Distillation via Epistemic Complexity Approximation

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Abstract

Distillation of reasoning chains from Large Language Models (LLMs) into their smaller siblings is a standard training approach. Quite frequently researchers do not have full access to a teacher model and its internal representations, which limits the resulting performance. We propose a novel blueprint for efficient offline distillation that recursively uses lightweight epistemic and aleatoric complexity approximations for selective distillation. Specifically, for three open models (3-4B) at each epoch we estimate offline data complexity with a single-token answer entropy and use a medium model’s (70B) entropy as an aleatoric proxy. Next, we apply selective distillation only for a small subset of the reasoning chains with the highest corresponding epistemic component. Our approach consistently outperforms naive distillation on comparable computational budget (57.15 ± 1.01 vs 51.61 on average) and offers a much steeper learning curve (matching baseline score on 15% of the data). Additionally, we present comparative analysis of methods to generate, normalize and augment synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain. We publish our code and data to facilitate further research in this direction.

1 Introduction

Modern LLMs come in different sizes, and their performance is often correlated with the number of parameters. The largest of them (GLM-5-Team et al., 2026; Kimi-Team et al., 2026; DeepSeek-AI et al., 2026; Qwen Team, 2026) demonstrate exceptional results on a vast variety of tasks and span 1-2 trillion parameters. Further domain adaptation (Cheng et al., 2024; Lu et al., 2024) or even inference at this scale requires massive computational resources (Wan et al., 2024) unavailable to most practitioners. It prompts the need for capable smaller models (Chen and Varoquaux, 2026)

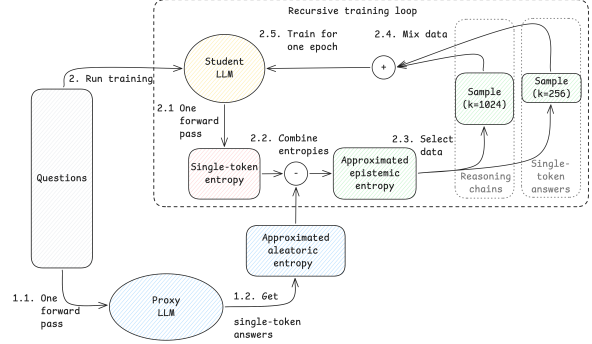


Figure 1: Recursive offline distillation scheme

that can be fine-tuned and deployed in resource-constrained environments.

One of the most prominent approaches for small model training is distillation from a larger teacher model (Hinton et al., 2015; Sanh et al., 2020; Xu et al., 2024). Many methods rely on having access to the online teacher, its logits and internal representations (Agarwal et al., 2024; Jiao et al., 2020; Gu et al., 2024), which is not always possible (Zhou et al., 2023; Ryskulov et al., 2026). It forces practitioners to rely on offline distillation, where the teacher model generates reasoning text for a given dataset, and the student model is trained on this data (Li et al., 2023; Hsieh et al., 2023).

Recent work has improved LLM distillation through training objectives (Ko et al., 2024; Xu et al., 2026). Alternatively, data curation (Just et al., 2026; Li et al., 2025; Yang et al., 2026) and curriculum learning (Jiang et al., 2025; Zhang and Liu, 2026) have been explored with the scoring criteria remaining a key challenge (Jia et al., 2026).

We propose a fully automated recursive pipeline for offline distillation (Figure 1) based on uncertainty decomposition (Wang et al., 2025): (1) measure the medium model’s entropy once, using it as the aleatoric uncertainty proxy; (2) for each training epoch, estimate the epistemic component, sample the dataset, and run the distillation on the

selected data mix. The method consistently outperforms naive distillation (57.15 ± 1.01 vs 51.61 on average), and it is designed to work in resource-constrained environments.

We validate our approach by fine-tuning three open-source models Qwen2.5 3B (Team, 2024), Phi-4-Mini (Microsoft et al., 2025), LLaMA 3.2 3B (Grattafiori et al., 2024) on a multiple-choice question answering dataset MMLU-Pro (Wang et al., 2024). To confirm how the method works in practice under hardware constraints, we run the experiments on a single A100/H100¹ GPU with 80GB of VRAM.

Additionally, we provide a comparative analysis of augmentation strategies for synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain and conduct sensitivity studies of the synthetic trace generation and normalization methods and sampling strategies.

2 Related Work

3 Method

4 Results

4.1 Experimental Setup

4.2 Main Results

4.3 Synthetic Traces Generation

4.4 Ablation Study

5 Conclusion and discussion

Limitations

References

- Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. 2024. [On-policy distillation of language models: Learning from self-generated mistakes](#). In *International Conference on Learning Representations*, volume 2024, pages 21246–21263.
- Lihu Chen and Gaël Varoquaux. 2026. [What is the role of small models in the llm era: A survey](#). *Preprint*, arXiv:2409.06857.
- Daixuan Cheng, Shaohan Huang, and Furu Wei. 2024. [Adapting large language models via reading comprehension](#). In *International Conference on Learning Representations*, volume 2024, pages 48624–48652.
- DeepSeek-AI, Anyi Xu, Bangcai Lin, Bing Xue, Bingxuan Wang, Bingzheng Xu, Bochao Wu, Bowei Zhang, Chaofan Lin, Chen Dong, Chenchen Ling, Chengda

- Lu, Chenggang Zhao, Chengqi Deng, Chengyu Hou, Chenhao Xu, Chenze Shao, Chong Ruan, Conner Sun, and 300 others. 2026. [Deepseek-v4: Towards highly efficient million-token context intelligence](#). *Preprint*, arXiv:2606.19348.

- GLM-5-Team, :, Aohan Zeng, Xin Lv, Zhenyu Hou, Zhengxiao Du, Qinkai Zheng, Bin Chen, Da Yin, Chendi Ge, Chenghua Huang, Chengxing Xie, Chenzheng Zhu, Congfeng Yin, Cunxiang Wang, Gengzheng Pan, Hao Zeng, Haoke Zhang, Hao-ran Wang, and 168 others. 2026. [Glm-5: from vibe coding to agentic engineering](#). *Preprint*, arXiv:2602.15763.

- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. [The llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.

- Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. 2024. [Minillm: Knowledge distillation of large language models](#). In *International Conference on Learning Representations*, volume 2024, pages 32694–32717.

- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. [Distilling the knowledge in a neural network](#). *Preprint*, arXiv:1503.02531.

- Cheng-Yu Hsieh, Chun-Liang Li, Chih-kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. [Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017, Toronto, Canada. Association for Computational Linguistics.

- Yaning Jia, Chunhui Zhang, Xingjian Diao, Xiangchi Yuan, Zhongyu Ouyang, Chiyu Ma, and Soroush Vosoughi. 2026. [What makes a good curriculum? disentangling the effects of data ordering on LLM mathematical reasoning](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 34472–34488, San Diego, California, United States. Association for Computational Linguistics.

- Wangyi Jiang, Yaojie Lu, Hongyu Lin, Xianpei Han, and Le Sun. 2025. [Teach small models to reason by curriculum distillation](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 7412–7422, Suzhou, China. Association for Computational Linguistics.

- Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. 2020. [TinyBERT: Distilling BERT for natural language understanding](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4163–

¹We had to switch cards depending on availability

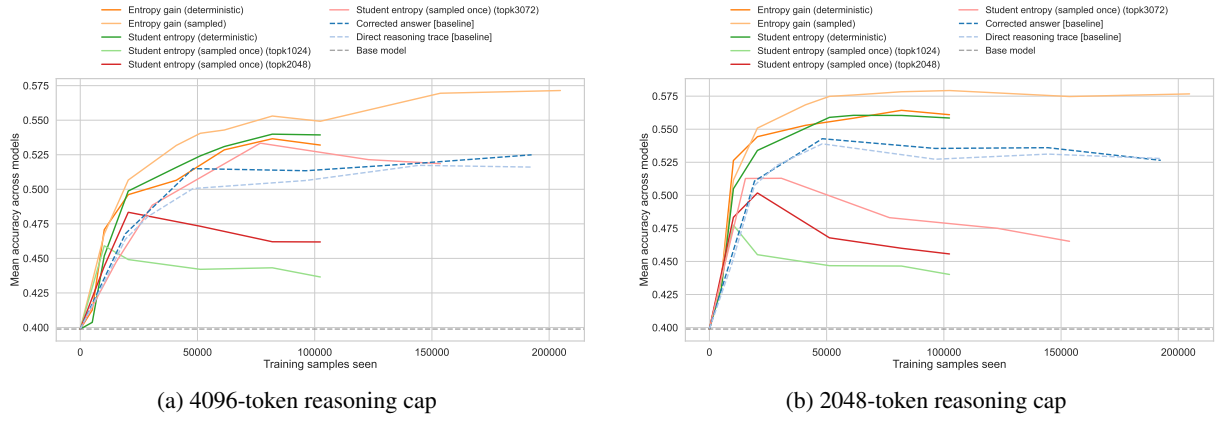


Figure 2: Distillation pipeline results across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro

Method	LLaMA-3B	Qwen-3B	Phi4-mini	Mean
Direct reasoning trace	46.09 / 45.55	51.45 / 50.37	57.27 / 62.43	51.61 / 52.78
Corrected answer	47.80 / 47.26	50.87 / 49.42	58.81 / 61.26	52.49 / 52.65
Student entropy (deterministic)	49.83 / 50.08	53.62 / 53.03	58.35 / 64.42	53.93 / 55.85
Entropy gain (deterministic)	<u>50.17 / 50.37</u>	51.04 / <u>53.28</u>	58.40 / 64.63	53.20 / <u>56.10</u>
Student entropy (sampled once, $k = 1024$)	41.35 / 40.11	39.44 / 39.07	50.17 / 52.87	43.65 / 44.01
Student entropy (sampled once, $k = 2048$)	42.19 / 41.06	43.18 / 43.02	53.20 / 52.62	46.19 / 45.57
Student entropy (sampled once, $k = 3072$)	42.02 / 42.81	56.77 / 40.86	56.77 / 55.90	51.86 / 46.52
Entropy gain (sampled)	52.58 $\pm$ 1.94 /	56.64 $\pm$ 3.79 /	62.22 $\pm$ 2.42 /	57.15 $\pm$ 1.01 /
	52.08 $\pm$ 2.28	56.86 $\pm$ 2.32	64.07 $\pm$ 4.52	57.67 $\pm$ 1.19

Table 1: Distillation pipeline last epoch accuracy (%) across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro, reported for 4096-token / 2,048-token thinking budget. Sampled entropy gain uses three seeds per model and includes $\pm$ 95% confidence-interval. Other methods are deterministic.

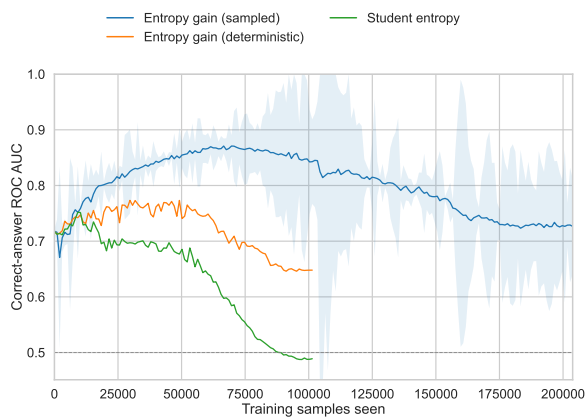


Figure 3: Student calibration

4174, Online. Association for Computational Linguistics.

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Hoang Anh Just, Myeongseob Ko, and Ruoxi Jia. 2026. [The signal is in the steps: Local scoring for reasoning data selection](#). In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 55125–55155. PMLR.

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Kimi-Team, Tongtong Bai, Yifan Bai, Yiping Bao, M. C., Jianfeng Cai, Xinyuan Cai, Peizhou Cao, Yuxuan Cao, Ziwei Chai, Y. Charles, H. S. Che, Guanduo Chen, Guangyu Chen, Guanzheng Chen, Huarong Chen, Jia Chen, Jianlong Chen, Jun Chen, and 383 others. 2026. [Kimi k3: Open frontier intelligence](#). *Preprint*, arXiv:2607.24653.

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Jongwoo Ko, Sungnyun Kim, Tianyi Chen, and Se-Young Yun. 2024. [DistiLLM: Towards streamlined distillation for large language models](#). In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 24872–24895. PMLR.

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Liunian Harold Li, Jack Hessel, Youngjae Yu, Xiang Ren, Kai-Wei Chang, and Yejin Choi. 2023. [Symbolic chain-of-thought distillation: Small models can also “think” step-by-step](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2665–

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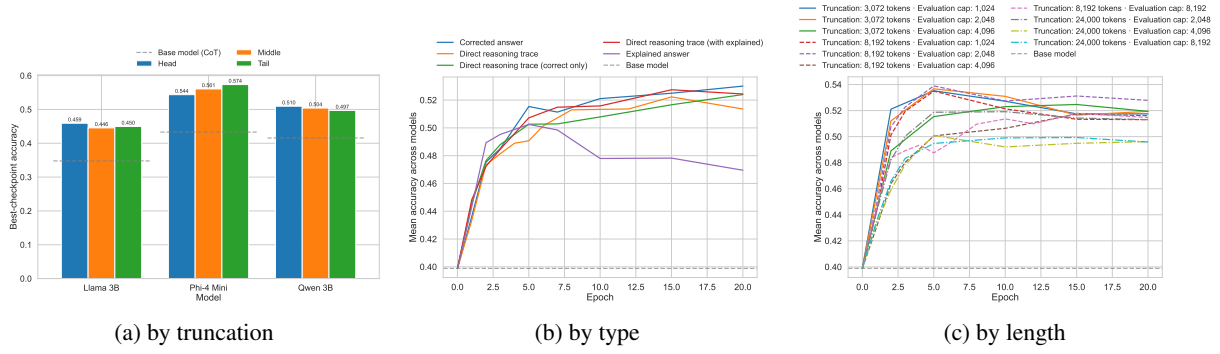


Figure 4: Comparison of synthetic traces

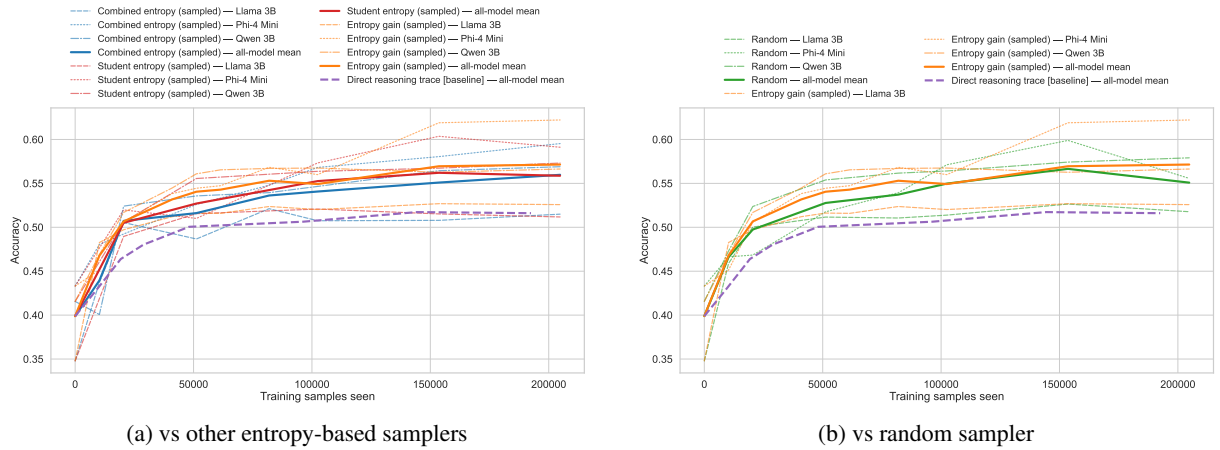


Figure 5: Comparison of entropy gain sampler

2679, Toronto, Canada. Association for Computational Linguistics.

Yang Li, Youssef Emad, Karthik Padthe, Jack Lanchantin, Weizhe Yuan, Thao Nguyen, Jason Weston, Shang-Wen Li, Dong Wang, Ilia Kulikov, and Xian Li. 2025. *Naturalthoughts: Selecting and distilling reasoning traces for general reasoning tasks*. *Preprint*, arXiv:2507.01921.

Wei Lu, Rachel K. Luu, and Markus J. Buehler. 2024. *Fine-tuning large language models for domain adaptation: Exploration of training strategies, scaling, model merging and synergistic capabilities*. *Preprint*, arXiv:2409.03444.

Microsoft, :, Abdelrahman Abouelenin, Atabak Ashfaq, Adam Atkinson, Hany Awadalla, Nguyen Bach, Jianmin Bao, Alon Benhaim, Martin Cai, Vishrav Chaudhary, Congcong Chen, Dong Chen, Dongdong Chen, Junkun Chen, Weizhu Chen, Yen-Chun Chen, Yi ling Chen, Qi Dai, and 57 others. 2025. *Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras*. *Preprint*, arXiv:2503.01743.

Qwen Team. 2026. *Qwen3.8-max: A new bar for coding and cowork*.

Bakbergen Ryskulov, Iker García-Ferrero, David Montero, David Jansen, Ali Hashemi, Jezabel R. Gar-

cia, Antonio Tiene, and Román Orús. 2026. *Efficient knowledge distillation for llms: Offline top-k logits and a fused chunked kl loss*. *Preprint*, arXiv:2608.03796.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2020. *Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter*. *Preprint*, arXiv:1910.01108.

Qwen Team. 2024. *Qwen2.5: A party of foundation models*.

Zhongwei Wan, Xin Wang, Che Liu, Samiul Alam, Yu Zheng, Jiachen Liu, Zhongnan Qu, Shen Yan, Yi Zhu, Quanlu Zhang, Mosharaf Chowdhury, and Mi Zhang. 2024. *Efficient large language models: A survey*. *Preprint*, arXiv:2312.03863.

Tianyang Wang, Yunze Wang, Jun Zhou, Benji Peng, Xinyuan Song, Charles Zhang, Xintian Sun, Qian Niu, Junyu Liu, Silin Chen, Keyu Chen, Ming Li, Pohsun Feng, Ziqian Bi, Ming Liu, Yichao Zhang, Cheng Fei, Caitlyn Heqi Yin, and Lawrence KQ Yan. 2025. *From aleatoric to epistemic: Exploring uncertainty quantification techniques in artificial intelligence*. *Preprint*, arXiv:2501.03282.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max

Sampler	LLaMA-3B	Qwen-3B	Phi4-mini	Mean
Random	51.77 $\pm$ 2.76	57.91 $\pm$ 1.40	55.57 $\pm$ 2.54	55.08 $\pm$ 0.80
Student entropy	51.18 $\pm$ 3.11	57.32 $\pm$ 1.52	59.12 $\pm$ 5.22	55.87 $\pm$ 1.40
Combined entropy	51.50	56.90	59.53 $\pm$ 2.11	55.98
Entropy gain	52.58 $\pm$ 1.94	56.64 $\pm$ 3.79	62.22 $\pm$ 2.42	57.15 $\pm$ 1.01

Table 2: Distillation pipeline last epoch accuracy for different sampling methods across three models (Qwen-3B, LLaMA-3B, and Phi4-mini) on MMLU-Pro, reported for 4096-token thinking budget. Each method is evaluated with 3 seeds per model and includes $\pm$ 95% confidence-interval.

Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. [Mmlu-pro: A more robust and challenging multi-task language understanding benchmark](#). *Preprint*, arXiv:2406.01574.

Shuyao Xu, Cheng Peng, Jiangxuan Long, Weidi Xu, Wei Chu, and Yuan Qi. 2026. [Harnessing negative signals: Reinforcement distillation from teacher data for LLM reasoning](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1618–1639, San Diego, California, United States. Association for Computational Linguistics.

Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao, and Tianyi Zhou. 2024. [A survey on knowledge distillation of large language models](#). *Preprint*, arXiv:2402.13116.

Cehao Yang, Xueyuan Lin, Xiaojun Wu, Chengjin Xu, Xuhui Jiang, Honghao Liu, Hui Xiong, and Jian Guo. 2026. [Select2Reason: Efficient instruction-tuning data selection for long-CoT reasoning](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 6656–6671, San Diego, California, United States. Association for Computational Linguistics.

Mengxiang Zhang and Lingyuan Liu. 2026. [Calibrated progressive distillation: Co-designing curriculum and target mixing for knowledge distillation of large language models](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 6757–6776, San Diego, California, United States. Association for Computational Linguistics.

Qinhong Zhou, Zonghan Yang, Peng Li, and Yang Liu. 2023. [Bridging the gap between decision and logits in decision-based knowledge distillation for pre-trained language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13234–13248, Toronto, Canada. Association for Computational Linguistics.

A Example Appendix

This is an appendix.